To bind or not to bind? Replaying Object-centric Processes under stable Relationships

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Abstract. Object-centric event logs record the intertwined behavior of multiple business objects. To replay these event logs, object-centric Petri nets (OCPN) have been introduced. While OCPNs indicate how objects of different types flow through the process and how they co-participate in events, they do not provide mechanisms for tracking the identity of objects and their relationships, making the synchronization of objects ambiguous. Yet, OCPNs often appear informally "just fine" in intuitively capturing the intended process semantics for a large number of object-centric executions, whose characteristics have never been made explicit. The ambiguity of OCPNs can be resolved in models that provide explicit support for object identifiers, such as object-centric Petri nets with identifiers (OPIDs). In this paper, we aim to close the gap between OCPNs and OPIDs rigorously by formally identifying the class of object-centric executions with stable many-to-one relationships. They yield a non-ambiguous interpretation of synchronization semantics in a given OCPN. We then show that every OCPN can be unambiguously encoded into a corresponding OPID where objects, stable relationships, and synchronization points are made explicit so that the two specifications coincide when focusing on executions with stable relationships.

Keywords: Business Process Management · Conformance Checking · Object-centric Petri Nets · Object-centric Event Logs · Synchronization.

1 Introduction

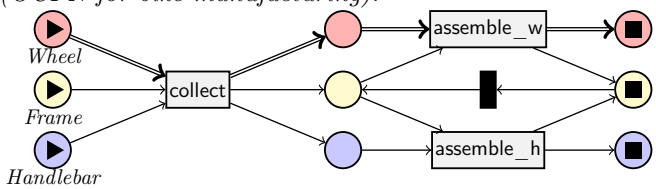
Process mining analyzes behavior in information systems through event logs to gain insights into the system's behavior [3]. This differs from general data science techniques not only because of the specificity of the input data but also because operating with white-box (process) models is usually a strong requirement. Process mining techniques strive between two contrasting forces when dealing with process models: capturing complex real-life processes demands *expressiveness*, while performance and memory consumption of the adopted algorithms call for *simplicity*. Balancing these two forces is even more challenging in light of the fact that within a process mining pipeline, the same model may carry a descriptive or prescriptive nature, depending on the context. For example, a process model automatically discovered from an event log is *descriptive* for the input event data but may later be used for conformance checking [7], where executions can be recognized by the *prescriptive* model or not.

Achieving a suitable trade-off appears particularly difficult within the novel research area of object-centric process science, where the standard assumption of case-centricity is removed, and processes at once capture multiple, co-evolving, interrelated objects that represent different case perspectives [1]. At the event data level, the minimal requirement is to record which objects are referred by events and the types of such objects. Different representations are being explored in this spectrum [8], most prominently *object-centric event logs* (OCEL [9]). For example, consider the following OCEL L_1 that represents a recorded behavior in a bicycle manufacturing workshop. For a given order, a frame, a set of wheels, and a handlebar are collected before being assembled:

$$\begin{aligned} L_1 = [\ & \langle \mathsf{collect}, \{f_1, h_1, w_1, w_2\}\rangle, \langle \mathsf{assemble_w}, \{f_1, w_1, w_2\}\rangle, \\ & \langle \mathsf{assemble_h}, \{f_1, h_1\}\rangle, \langle \mathsf{collect}, \{f_2, h_2, w_3, w_4\}\rangle, \\ & \langle \mathsf{assemble_h}, \{f_2, h_2\}\rangle, \langle \mathsf{assemble_w}, \{f_2, w_3, w_4\}\rangle \quad] \end{aligned}$$

with f_1 of type *frame*, h_1 of type *handlebar*, and w_i of type *wheel*. To replay and describe the behavior captured in an OCEL, an object-centric Petri net (OCPN) can be automatically discovered [4]. For example, the OCPN in Example 1 allows for replaying the behavior of log L_1. In the initial marking (▶), all recorded objects of the log are placed. The transitions can fire by binding objects and processing them through the net. A non-variable arc binds a single object, while a variable arc (double line) binds a set of objects. Finally, the net accepts a given execution if all objects are placed in the final marking (■).

Example 1 (OCPN for bike manufacturing).



The tension between simplicity and expressiveness emerges to its full extent: while OCPNs indicate how objects of different types flow through the process and how they co-participate in process activities, they do not provide mechanisms for tracking the identity of objects and their relationships. As pointed out in [2], this deliberate underspecification makes discovered OCPNs underfitting, as they allow for too much behavior. In Example 1, this makes the synchronization of objects ambiguous: wheels collected for a given frame may be *reshuffled*, that is, assembled to different frames, which is undesired. For example, the example OCPN accepts the execution in the following log:

$$\begin{aligned} L_2 = \ & [\langle \mathsf{collect}, \{f_3, h_3, w_5, w_6\}\rangle, \langle \mathsf{collect}, \{f_4, h_4, w_7\}\rangle, \langle \mathsf{assemble_w}, \{f_3, w_5\}\rangle, \\ & \langle \mathsf{assemble_h}, \{f_3, h_4\}\rangle, \langle \mathsf{assemble_w}, \{f_4, w_6, w_7\}\rangle, \langle \mathsf{assemble_h}, \{f_4, h_3\}\rangle] \end{aligned}$$

where wheel 6 is attached to frame 4 even though it was related to frame 3 at collection time. This indicates that whenever an OCPN is used in a prescriptive

manner for conformance checking or simulation, further so-called "bounding behavior" [2] is needed restricting to only desired synchronizations. Such a bounding behavior can be incorporated into the process model by resorting to Petri nets with identifiers (PNIDs), which provide explicit modeling mechanisms for manipulating objects and their relationships [10, 18]. Specifically, *object-centric PNIDs* (OPIDs [11]) incorporate the notion of variable arc from OCPNs into PNIDs, thus providing a natural candidate for enriching an OCPN with the desired bounding behavior. In general, an OCPN can be encoded into different OPIDs, depending on the desired additional bounding behavior.

While moving from OCPNs to OPIDs increases expressiveness, it heavily impacts algorithmic techniques. Discovery [4] and conformance [15] of OCPNs can, in fact, be efficiently tackled using divide-and-conquer approaches that split the OCEL log into case-centric logs for different types, apply off-the-shelf conventional process mining techniques, and then modularly recombine the so-obtained results. This modularity does not carry over to OPIDs, as the identity of objects and their relationships requires global checks. For this reason, discovery of OPIDs is ongoing research, while conformance is handled in [11] through a monolithic approach. In addition, OCPNs often appear informally "just fine" in intuitively capturing the intended process semantics for a large number of object-centric executions. Yet their characteristics have never been made explicit. This leads to the two main research questions we aim to answer in this paper rigorously:

RQ1 Which executions are suitably described by an OCPN without the need to further specify its behavior?

RQ2 How can these executions of an OCPN be uniquely encoded into a corresponding OPID tracking object identities and relationships unambiguously?

We answer both questions affirmatively for executions that yield *stable relationships* for their involved objects: for a many-to-one relationship present in the process, whenever an object on the "many" side is bound to a corresponding object on the "one" side, the binding rigidly holds throughout the entire execution (thus making it impossible to cease the relationship and create a new binding later). This characteristic holds for L_1: the frame, its wheels, and its handlebar form stable relationships throughout the entire execution. The stability is instead violated by L_2, where $\langle \mathsf{collect}, \{f_3, h_3, w_5, w_6\}\rangle$ implicitly establishes the link (f_3, w_6), which is then violated by $\langle \mathsf{assemble_w}, \{f_4, w_6, w_7\}\rangle$.

The idea of focusing on stable relationships stems from the observation that many OCEL logs indeed satisfy the stability principle of L_1 for many-to-one relationships, and would not contain changing bindings as in L_2. This observation is further corroborated by considering that, in previous work on simulating OCPNs without further bounding behavior [13], many-to-one relationships extracted from the input OCEL are considered rigid. And even in the OCEL 2.0 standard [6], relationship links between objects do not change over time.

We substantiate these informal ideas through a formal approach consisting of four main steps. First, we consider arbitrary executions, showing how to map an OCPN into an equivalent OPID that retains the same ambiguity in dealing with object identity and relationships. Second, we formalize *stable many-to-one*

relationships on pairs of object types, formally capturing that whenever two objects of those types co-participate in an event, they form a stable relationship that is never altered during the execution. Third, we provide an encoding that, given an input OCPN and stable relationships, produces a corresponding OPID in which the identity of objects is tracked, and stable relationships are established and used in synchronization points. Finally, we show that, while the so-obtained OPID is more restrictive than the original OCPN, they are indeed identical when restricting only to executions with stable relationships.

The remainder of this paper is followed by preliminary formalisms in Section 2. Section 3 defines an equivalent mapping from OCPN to OPID, which is extended by Section 4 to implement stable many-to-one relationships. Section 5 discusses our contribution in the context of related work.

2 Preliminary Definitions

2.1 Object-centric event logs and object-centric Petri nets

To define OCELs and OCPNs, we recall notions by van der Aalst and Berti [4].

Definition 1. *Given a set of object types Σ, a set of activities $\mathcal{A}$, and a set of timestamps $\mathbb{T}$ under a total order $<$, an* object-centric event log *(OCEL) is a tuple $L = (E, \mathcal{O}, \pi_{act}, \pi_{obj}, \pi_{time})$ where:*

- *E is a set of events;*
- *$\mathcal{O}$ is a set of object identifiers, each typed by the function $ot : \mathcal{O} \to \Sigma$;*
- *the functions $\pi_{act} : E \to \mathcal{A}$, $\pi_{obj} : E \to \mathcal{P}(\mathcal{O})$, and $\pi_{time} : E \to \mathbb{T}$, mapping each event to an activity, a set of objects, and a timestamp.*

For example, the event logs L_1 and L_2 in Section 1 are defined for the object types *Wheel*, *Frame*, and *Handlebar*. The event $e = \langle \mathsf{collect}, \{f_1, h_1, w_1, w_2\}\rangle$ in L_1 refers to the activity $\pi_{act}(e) = \texttt{collect}$ and to the objects $\pi_{obj}(e) = \{f_1, h_1, w_1, w_2\}$, where $ot(f_1) = \mathit{Frame}$

Definition 2. *Given a set of object types Σ, and a set of activities $\mathcal{A}$, an* object-centric Petri net *(OCPN) is defined as a tuple $N = (P, T, F, pt, F_{var}, \ell)$, where:*

1. *P and T are finite sets of places and transitions such that $P \cap T = \emptyset$,*
2. *$F \subseteq (P \times T) \cup (T \times P)$ is the flow relation between places and transitions,*
3. *$pt\colon P \to \Sigma$ maps every place to an object type,*
4. *$F_{var} \subseteq F$ is the set of variable flow relations,*
5. *$\ell\colon T \to \mathcal{A} \cup \{\tau\}$ is the transition labelling where τ marks an invisible activity.*

The pre-places of a transition t are all places with ingoing flow relations $\bullet t = \{p \in P | (p, t) \in F\}$, while the post-places of t are all those connected with outgoing flow relations $t\bullet = \{p \in P | (t, p) \in F\}$. For a transition t, the function $tpl(t)$ returns the types of all its corresponding places $p \in \bullet t \cup t\bullet$. $tpl_{var}(t)$ returns all place types connected via variable flow, and $tpl_{nv}(t)$ all place

types connected via non-variable flow. An OCPN is considered well-formed if all flow relations of one transition agree in the variability for the place types, i.e., $\forall t \in T : tpl_{var}(t) \cap tpl_{nv}(t) = \emptyset$.

All possible tokens and their location in a place are defined as $Q = \{(p, o) \in P \times \mathcal{O} \mid pt(p) = type(o)\}$. A marking is generally defined as a multiset of token-object pairs $M\colon Q \to \mathbb{N}$ [4]. However, we pose the following specific assumptions about the structure of an OCPN. The contemporary discovery algorithms for OCPN utilize the Inductive Miner algorithm [14], which results in sound workflow nets for each object type. As a result, all places are bound to at most one object identity by design. Therefore, we interpret the marking M of an OCPN as a set of objects in a marking such that $M\colon Q \to \{0, 1\}$. Furthermore, each object type has one initial place and one final place.

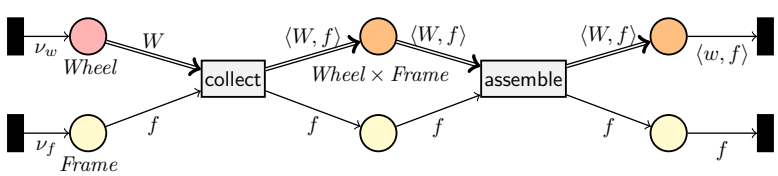
The set of binding executions is $B = \{(t, b) \in T \times (\Sigma \to \mathcal{P}(\mathcal{O})) \mid dom(b) = tpl(t) \wedge \forall ot \in tpl_{nv}(t) : |b(ot)| = 1\}$. It contains pairs of a transition t and a binding b. A binding b is a function that maps an object type $ot \in \Sigma$ to a set of objects. We denote by $codom(b)$ the codomain of b, i.e., $codom(b) = \bigcup_{\sigma \in \Sigma} b(\sigma)$. This binding execution consumes a multiset of object tokens from the transition's pre-places $cons(t, b) = [(p, o) \in Q \mid p \in \bullet t \wedge o \in b(pt(p))]$. Similarly, it produces object tokens $prod(t, b) = [(p, o) \in Q \mid p \in t \wedge \bullet\, o \in b(pt(p))]$ This binding is enabled in a marking M_i and can be executed if $cons(t, b) \leq M_i$. The resulting marking is $M_j = M_i - cons(t, b) + prod(t, b)$. The firing is denoted as $M_i \xrightarrow{(t,b)} M_j$. $B \subseteq T \times \mathcal{O}$ is the set of all possible bindings, such that a binding (t, b) for a transition t considers only objects $b \subseteq \mathcal{O}$ that belong to the types of the respective input places and are in line with the variable and non-variable arcs.

An accepting OCPN combines an OCPN N with an initial marking $M_{init} \in \mathcal{B}(Q)$ and a final marking $M_{final} \in \mathcal{B}(Q)$. An accepted run is a sequence of binding executions from the initial to the final marking $M_{init} \xrightarrow{*} M_{final}$

The accepting OCPN can be used to *replay* the behavior of the log. All recorded objects are interpreted as the initial marking M_{init}. An event in the log and its recorded objects are interpreted as a binding execution of the corresponding transition and the involved objects, which results in a new marking. After replaying all events consecutively, the OCPN can accept the log if the run results in the final marking M_{final}. If an event has no corresponding binding execution, or M_{final} is not reached, the log is rejected by the OCPN.

2.2 Object-centric Petri nets with identifiers

Example 2 (Simple OPID for bike manufacturing). The OPID combines the object types *Wheel* (red), *Frame* (yellow), and relations for both (orange places).



We summarize the definitions from [11]. Every object type $\sigma \in \Sigma$ is assumed to have a domain $dom(\sigma) \subseteq \mathcal{O}$, given by all objects in $\mathcal{O}$ of type σ. List types with a base type in σ are denoted as $[\sigma]$. Each Petri net place has a *color* that is a cartesian product of data types from Σ; precisely, the set of colors $\mathcal{C}$ is the set of all $\sigma_1 \times \cdots \times \sigma_m$ such that $m \geq 1$ and $\sigma_i \in \Sigma$ for all $1 \leq i \leq m$. We fix a set of Σ-typed variables $\mathcal{X} = X \uplus X_{list} \uplus \Upsilon$ as the disjoint union of a set X of "normal" variables that refer to single objects, denoted by lower-case letters like x, with a type $\text{type}(x) \in \Sigma$, a set X_{list} of list variables that refer to a list of objects of the same type, denoted by upper case letters like U, with a type $\text{type}(U) = [\sigma]$ for some $\sigma \in \Sigma$, and a set Υ of variables referring to fresh objects, denoted as ν, with $\text{type}(\nu) \in \Sigma$.

In OPIDs, tokens are object tuples associated with a color. To define relationships between objects in consumed and produced tokens when firing a transition, arc *inscriptions* are defined.

Definition 3. *An* inscription *is a tuple* $\boldsymbol{x} = \langle x_1, \ldots, x_m \rangle$ *such that* $m \geq 1$ *and* $x_i \in \mathcal{X}$ *for all* i*, but at most one* $x_i \in X_{list}$*, for* $1 \leq i \leq m$*. We call* $\boldsymbol{x}$ *a* template inscription *if* $x_i \in X_{list}$ *for some* i*, and a* simple inscription *otherwise.*

Template inscriptions capture an arbitrary number of tokens of the same color; they are used to label the correspondent of variable arcs in OCPNs. The color of an inscription $\iota = \langle x_1, \ldots, x_m \rangle$ is given by $\text{color}(\iota) = \langle \sigma_1, \ldots, \sigma_m \rangle$ where $\sigma_i = \text{type}(x_i)$ if $x_i \in X \cup \Upsilon$, and $\sigma_i = \sigma'$ if x_i is a list variable of type $[\sigma']$. We set $\text{vars}(\iota) = \{x_1, \ldots, x_m\}$. The set of all inscriptions is denoted Ω.

Definition 4. *An* object-centric Petri net with identifiers *(OPID) is defined as a tuple* $N = (\Sigma, P, T, F_{in}, F_{out}, \text{color}, \ell)$*, where:*

1. P *and* T *are finite sets of places and transitions such that* $P \cap T = \emptyset$*,*
2. $\text{color}\colon P \to \mathcal{C}$ *maps every place to a color,*
3. $\ell\colon T \to \mathcal{A} \cup \{\tau\}$ *is the transition labelling where* τ *marks an invisible activity,*
4. $F_{in}\colon P \times T \to \Omega$ *is a partial function called* input flow*, such that* $\text{color}(F_{in}(p,t)) = \text{color}(p)$ *for every* $(p,t) \in dom(F_{in})$*,*
5. $F_{out}\colon T \times P \to \Omega$ *is a partial function called* output flow*, such that* $\text{color}(F_{out}(t,p)) = \text{color}(p)$ *for every* $(t,p) \in dom(F_{out})$*.*

We set $\text{vars}_{in}(t) = \bigcup_{p \in P} \text{vars}(F_{in}(p,t))$*, and* $\text{vars}_{out}(t) = \bigcup_{p \in P} \text{vars}(F_{out}(t,p))$*, and require that* $\text{vars}_{in}(t) \cap \Upsilon = \emptyset$ *and* $\text{vars}_{out}(t) \subseteq \text{vars}_{in}(t) \cup \Upsilon$*, for all* $t \in T$*.*

If $F_{in}(p,t)$ is defined, it is called a *variable* flow if $F_{in}(p,t)$ is a template inscription, and *non-variable* flow otherwise; and similar for output flows. Variable flows play the role of variable arcs in OCPNs [4]; they can carry multiple tokens at once. The common notations $\bullet t = \{p \mid (p,t) \in dom(F_{in})\}$ and $t\bullet = \{p \mid (t,p) \in dom(F_{out})\}$ are used as well.

Semantics. Given the set of objects $\mathcal{O}$, the set of *tokens* $\boldsymbol{\mathcal{O}}$ is the set of object tuples $\boldsymbol{\mathcal{O}} = \{\mathcal{O}^m \mid m \geq 1\}$. The *color* of a token $\omega \in \boldsymbol{\mathcal{O}}$ of the form $\omega = \langle o_1, \ldots, o_m \rangle$ is denoted $\text{color}(\omega) = \langle \text{type}(o_1), \ldots, \text{type}(o_m) \rangle$. A *marking* of an OPID $N = \langle \Sigma, P, T, F_{in}, F_{out}, \text{color}, \ell \rangle$ is a function $M\colon P \to 2^{\boldsymbol{\mathcal{O}}}$, such that for

all $p \in P$ and $\langle o_1, \dots o_m \rangle \in M(p)$, it holds that $\text{color}(\langle o_1, \dots o_m \rangle) = \text{color}(p)$. Let $\mathit{Lists}(\mathcal{O})$ denote the set of objects lists of the form $[o_1, \dots, o_k]$ with $o_1, \dots, o_k \in \mathcal{O}$ such that all o_i have the same type; the type of such a list is then $[\text{type}(o_1)]$. Next, *bindings* are defined to fix which objects are involved in a transition firing.

Definition 5. *A* binding *for a transition t and a marking M is a type-preserving function $\beta\colon \text{vars}_{in}(t) \cup \text{vars}_{out}(t) \to \mathcal{O} \cup \mathit{Lists}(\mathcal{O})$. To ensure the freshness of created values, β is required to be injective on $\Upsilon \cap \text{vars}_{out}(t)$, and $\beta(\nu)$ must not occur in M for all $\nu \in \Upsilon \cap \text{vars}_{out}(t)$.*

We denote by $\mathit{codom}(\beta) \subseteq \mathcal{O}$ the set of all objects occurring in tuples returned by β. Bindings are extended to inscriptions, denoted $\boldsymbol{\beta}$, to fix which tokens participate in a transition firing. For an inscription $\iota = \langle x_1, \dots, x_m \rangle$, let $o_i = \beta(x_i)$ for all $1 \leq i \leq m$. Then $\boldsymbol{\beta}(\iota)$ is the set of object tuples defined as follows: if ι is a simple inscription, then $\boldsymbol{\beta}(\iota) = \{\langle o_1, \dots, o_m \rangle\}$. Otherwise, there must be one x_i, $1 \leq i \leq n$, such that $x_i \in X_{list}$, and consequently o_i must be a list, say $o_i = [u_1, \dots, u_k]$ for some $u_1, \dots, u_k$. Then $\boldsymbol{\beta}(\iota) = \{\langle o_1, \dots, o_{i-1}, u_1, o_{i+1}, \dots, o_m \rangle, \dots, \langle o_1, \dots, o_{i-1}, u_k, o_{i+1}, \dots, o_m \rangle\}$. $\mathcal{B}$ denotes the set of all bindings. Finally, a transition with a binding β is enabled if all object tuples pointed by $\boldsymbol{\beta}$ occur in the current marking:

Definition 6. *A transition $t \in T$ and a binding β for marking M are* enabled *in M if $\boldsymbol{\beta}(F_{in}(p,t)) \subseteq M(p)$ for all $p \in \bullet t$.*

Definition 7. *Let transition t be enabled in marking M with binding β. The* firing *of t yields the new marking M' given by $M'(p) = M(p) \setminus \boldsymbol{\beta}(F_{in}(p,t))$ for all $p \in \bullet t \setminus t\bullet$, and $M'(p) = M(p) \cup \boldsymbol{\beta}(F_{out}(p,t))$ for all $p \in t\bullet \setminus \bullet t$, and $M'(p) = M(p) \setminus \boldsymbol{\beta}(F_{in}(p,t)) \cup \boldsymbol{\beta}(F_{out}(p,t))$ for all $p \in \bullet t \cap t\bullet$.*[1]

We write $M \xrightarrow{t,\beta} M'$ to denote that t is enabled for binding β in M, and its firing yields M'. An *accepting* OPID N is an OPID together with a set of initial markings M_{init} and a set of final markings M_{final}. A run is accepted if it starts in the initial and ends in the final marking $M_{init} \xrightarrow{*} M_{final}$.

3 Equivalent Object-centric Petri Nets with Identifiers

This section defines the mapping $\mathbb{T}_1$ from an OCPN N to an OPID. Second, we show that the replay of both nets N and $\mathbb{T}_1(N)$ on a given event log are equivalent, i.e., they accept and reject the same executions.

In this paper, we assume that for any given OCPN $N = (P, T, F, pt, F_{var}, \ell)$ there are fixed sets of places $P_{\blacktriangleright}, P_{\blacksquare} \subseteq P$ called *play* and *stop places*, respectively, such that for each $\sigma \in \Sigma$ there is exactly one place $p \in P_{\blacktriangleright}$ and $p' \in P_{\blacksquare}$ such that $pt(p) = pt(p') = \sigma$. These are the places that are supposed to be populated by initial and final markings, respectively. Markings M_{init} and M_{final} are *suitable* initial and final markings for an OCPN with play and stop places $P_{\blacktriangleright}, P_{\blacksquare}$ if they

[1] The case for $p \in \bullet t \cap t\bullet$ was omitted in [11], which is insufficient for cyclic nets.

put tokens only in those places, i.e., for all $o \in \mathcal{O}$, $M_{init}(p, o) = 0$ for all $p \notin P_{\blacktriangleright}$ and $M_{init}(p', o) = 0$ for all $p' \notin P_{\blacksquare}$, and similar for M_{final}.

Note that a given set of objects uniquely determines suitable initial and final markings for N: precisely, for a finite set of objects $O \subseteq \mathcal{O}$, let M^O_{init} (resp. M^O_{final}) be the marking such that for each $\sigma \in \Sigma$ and and each $o \in O_\sigma$, $M^O_{init}(p_\sigma, o) = 1$ (resp. $M^O_{final}(p_\sigma, o) = 1$) where p_σ is the unique place in $P_{\blacktriangleright}$ (resp. $P_{\blacksquare}$) of type σ. Our mapping $\mathbb{T}_1$ maps an OCPN N with start and stop places to an OPID $\mathbb{T}_1(N)$ which keeps all places and transitions of N. The flow relations are adjusted and extended with arc inscriptions. Moreover, $\mathbb{T}_1(N)$ has additional transitions for creating and removing objects. In this way, the OPID can, on-demand, populate the places in $P_{\blacktriangleright}$ with objects in an initial marking for N and consume objects from places in $P_{\blacksquare}$ once a final marking for N is reached. The initial and final markings for $\mathbb{T}_1(N)$ are always empty.

Precisely, the OPID has an emitting transition for every object type, i.e., we set $T_e := \{t_{e,\sigma} \mid \sigma \in \Sigma\}$; and a consuming transition for every object type i.e., we set $T_c := \{t_{c,\sigma} \mid \sigma \in \Sigma\}$. With its emitting transitions T_e, an OPID allows for creating arbitrary new objects. However, under the replay of an OCEL L, this set of objects is fixed and denoted as $\mathcal{O}_L$. Hence, when replaying the log L, the inscription variables $\nu_i \in F_{out}(T_e)$ will be bound exclusively to objects in $\mathcal{O}_L$. To be able to define suitable inscriptions, we assume that for every object type $\sigma \in \Sigma$ there is a normal variable $x_\sigma \in X$, a list variable $X_\sigma \in X_{list}$, and a fresh object variable $\nu_\sigma \in \Upsilon$. We then set:

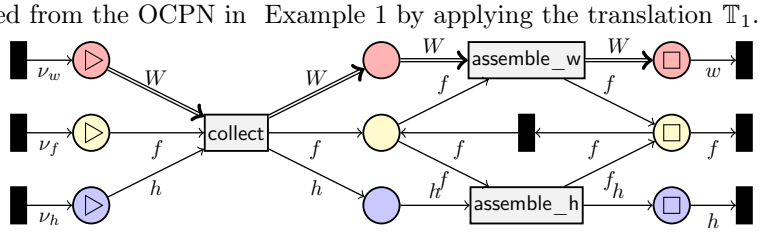
$$F_{out} \colon (T \cup T_e) \times P \to \Omega \qquad F_{out}(t,p) = \begin{cases} \langle x_{pt(p)} \rangle & \text{if } (t,p) \in F \setminus F_{var} \\ \langle X_{pt(p)} \rangle & \text{if } (t,p) \in F_{var} \\ \langle \nu_{pt(p)} \rangle & \text{if } p \in P_{\blacktriangleright} \wedge t \in T_e \end{cases}$$

$$F_{in} \colon P \times (T \cup T_c) \to \Omega \qquad F_{in}(p,t) = \begin{cases} \langle x_{pt(p)} \rangle & \text{if } (p,t) \in F \setminus F_{var} \\ \langle X_{pt(p)} \rangle & \text{if } (p,t) \in F_{var} \\ \langle x_{pt(p)} \rangle, & \text{if } p \in P_{\blacksquare} \wedge t \in T_c \end{cases}$$

The equivalent OPID is given by $\mathbb{T}_1(N) = (\Sigma_1, P_1, T_1, F_{in}, F_{out}, \mathit{color}, \ell_1)$ where $\Sigma_1 = \Sigma$, $P_1 = P$, $T_1 = T \cup T_e \cup T_c$, $\mathit{color}(p) = \langle pt(p) \rangle$ for all $p \in P$, and

$$\ell_1 \colon T \to \mathcal{A} \qquad \ell_1(t) = \begin{cases} \tau, & \text{if } t \in T_e \cup T_d \\ \ell(t), & \text{otherwise} \end{cases}$$

Example 3 (Equivalent OPID for bike manufacturing). The following OPID is obtained from the OCPN in Example 1 by applying the translation $\mathbb{T}_1$.



$\mathbb{T}_1(N)$ is interpreted as an accepting OPID where the initial and final markings are defined as the empty marking $M_\emptyset$, which requires all places to be empty.

The example OPID can replay the logs L_1 as well as L_2 as it has equivalent semantics to Example 1 in combining objects without additional constraints. The execution semantics of the OCPN N and the mapped OPID $\mathbb{T}_1(N)$ are equivalent by the definition of binding executions. Considering $M'_i = \mathbb{T}_1(M_i)$ as the mapped markings and $\beta_i = \mathbb{T}_1(b_i)$ as the mapped bindings, $\mathbb{T}_1(N)$ can produce and consume the same set of objects $\mathcal{O}$ as specified in the initial and final marking of the OCPN. For reasons of space, the proof of the following Theorem 1 was moved to Appendix A of the extended version of this paper.[2]

Theorem 1. *Let $\mathcal{O}$ be a set of objects, N be an accepting OCPN, $N' = \mathbb{T}_1(N)$, and $M_\emptyset$ denote the empty marking of N'. There exists a run of N:*

$$M_0 \xrightarrow{(t_1,b_1)} M_1 \xrightarrow{(t_2,b_2)} \cdots \xrightarrow{(t_n,b_n)} M_n$$

with $M_0{=}M^{\mathcal{O}}_{init}$ and $M_n{=}M^{\mathcal{O}}_{final}$ if and only if there is a run of N' over the set of objects $\mathcal{O}$:

$$M_\emptyset \xrightarrow{*} M'_0 \xrightarrow{(t_1,\beta_1)} M'_1 \xrightarrow{(t_2,\beta_2)} \cdots \xrightarrow{(t_n,\beta_n)} M'_n \xrightarrow{*} M_\emptyset$$

such that $M'_i = \mathbb{T}_1(M_i)$ for all $0 \leq i \leq n$ and $\beta_i = \mathbb{T}_1(b_i)$ for all $1 \leq i \leq n$, such that $M_\emptyset \xrightarrow{*} M'_0$ is an emitting sequence, $M'_n \xrightarrow{*} M_\emptyset$ a collapsing sequence, and no emitting or collapsing transitions occur in between.

4 Synchronization for Stable Many-to-One Relationships

In this section, we first define stable many-to-one relationships for an object-centric process execution. For a given OCEL and a replaying OCPN, we then extend the mapping $\mathbb{T}_1$ from Section 3 to implement the expected synchronization for the expected stable many-to-one relationship. Finally, we show that any execution of the OCPN is accepted by the mapped OPID if it conforms to the stable many-to-one relationships, and vice versa.

In the following, let a pair (σ_m, σ_o) be an *object type relationship* if $\sigma_m \neq \sigma_o$. For instance, for the bike manufacturing example, a stable many-to-one relationship may exist for *(Wheel, Frame)*. Given an OCEL $L = (E, \mathcal{O}, \pi_{act}, \pi_{obj}, \pi_{time})$, we define the function $lo\colon \mathcal{O} \times \Sigma \to \mathcal{O}$, which given an object and an object type, returns all co-occurring objects of this type, i.e., $lo(o, \sigma) = \bigcup_{e \in E}\{o' \mid o, o' \in \pi_{obj}(e) \wedge ot(o') = \sigma\}$. For OCPNs and OPIDs, lo is defined accordingly over a run of the net and bindings.

Definition 8. *An object type relationship (σ_m, σ_o) is a* stable many-to-one relationship *in an event log L if for all objects $o_m, o_o \in \mathcal{O}$, where $ot(o_m) = \sigma_m$ and $ot(o_o) = \sigma_o$ it holds that $|lo(o_m, \sigma_o)| = 1$.*

[2] `https://anonymous.4open.science/r/To-bind-or-not-to-bind-43DD`

Similarly for OCPNs and OPIDs, (σ_m, σ_o) is a *stable many-to-one relationship* if the same holds for a run in the net: The objects o_m of the many-side σ_m in the stable relationship (σ_m, σ_o) are required to be linked to exactly one object o_o of the one-side σ_o. On the other hand, every o_o must have a non-empty set of linked objects o_m. For example, *(Wheel, Frame)* is a stable many-to-one relationship in the log L_1 from Section 1, but not in L_2. Note that the one-to-one relationship between handlebar and frame manifests as two stable many-to-one relationships *(Handlebar, Frame)* and *(Handlebar, Frame)*.

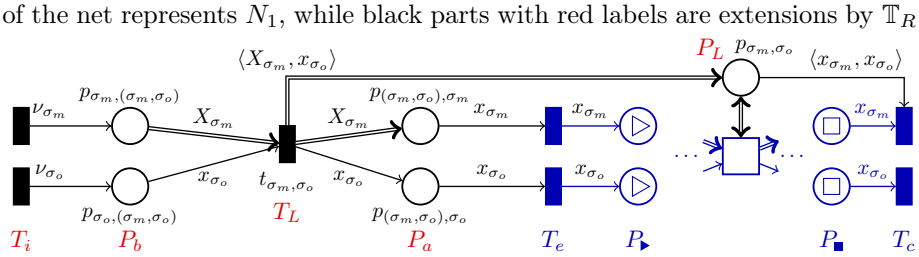
4.1 Mapping to synchronizing OPIDs

The goal of this mapping is to enforce a set of stable many-to-one relationships with the synchronization behavior of an OPID. To this end, for every many-to-one relationship (σ_m, σ_o), the required set of links $(m, o) \in \mathcal{O}^2$ of type (σ_m, σ_o) must be determined *before* the actual execution starts, and it must be stable during the execution. In general, the idea is to store all links of a relationship in a dedicated place, which is accessed by all transitions that operate on both object types. For the bike manufacturing Petri net, the translation results in the OPID shown in Example 4. For the log L_1 it determines the links between wheels w_1, w_2 of the type W and frame f_1 of type F as $(w_1, f_1), (w_2, f_1)$ before the execution starts and they are not changed. The transition's synchronization semantics ensure that only objects in a stable relationship are within one binding.

We assume a given OCPN $N = (P, T, F, pt, F_{var}, \ell)$, and a fixed set of object type relationships R, that are supposed to work as stable many-to-one relationships. The mapping $\mathbb{T}_R$ is defined on top of the mapping $\mathbb{T}_1$ in Section 3. Let $N_1 = \mathbb{T}_1(N) = (\Sigma_1, P_1, T_1, F_{in,1}, F_{out,1}, \text{color}_1, \ell_1)$. In $\mathbb{T}_R(N_1)$, we add the following new places and transitions:

- a set of *link places* $P_L := \{p_{(\sigma_m,\sigma_o)} \mid (\sigma_m, \sigma_o) \in R\}$, storing the links for each relationship $(\sigma_m, \sigma_o) \in R$, such that $\text{color}_2(p_{(\sigma_m,\sigma_o)}) = \langle \sigma_m, \sigma_o \rangle$,
- a set of *link transitions* $T_L := \{t_{\sigma_m,\sigma_o} \mid (\sigma_m, \sigma_o) \in R\}$ that creates links for a set of objects of type σ_m and one object of σ_o to populate P_L,
- new emitting transitions $T_i := \{t_{i,\sigma} \mid \exists \sigma' \in \Sigma : (\sigma, \sigma') \in R \vee (\sigma', \sigma) \in R\}$,
- a set $P_b := \{p_{\sigma,(\sigma_m,\sigma_o)} \mid (\sigma = \sigma_m \vee \sigma = \sigma_o) \wedge (\sigma_m, \sigma_o) \in R\}$ of auxiliary places before the link creation such that $\text{color}_2(p_{\sigma,(\sigma_m,\sigma_o)}) = \langle \sigma \rangle$, and
- a set $P_a := \{p_{(\sigma_m,\sigma_o),\sigma} \mid (\sigma = \sigma_m \vee \sigma = \sigma_o) \wedge (\sigma_m, \sigma_o) \in R\}$ of auxiliary places after link creation such that $\text{color}_2(p_{\sigma,(\sigma_m,\sigma_o)}) = \langle \sigma \rangle$.

The arrangement of these additional places and transitions is schematically shown below for one pair of object types $(\sigma_m, \sigma_o) \in R$. Here the blue part of the net represents N_1, while black parts with red labels are extensions by $\mathbb{T}_R$.



Before defining the new flow relations, we give an intuitive explanation: The idea is that all links are concurrently created before the objects are placed on the initial places $P_{\blacktriangleright}$ by T_e. To that end, new emitting transitions T_i are used to create objects which are placed in P_b. Every link creation transition $t_{\sigma_m,\sigma_o} \in T_L$ is connected with inflow to the according places from P_b, and with outflow to the respective link place $p_{\sigma_m,\sigma_o} \in P_L$ as well as places in P_a. The places in P_a are connected to the former emitting transitions T_e. When all transitions T_e emitted the objects to the places $P_{\blacktriangleright}$, the places in P_L are thus populated with a fixed set of links. Crucially, to enforce the synchronization for a transition t in N_1, which operates on σ_m and σ_o, the respective transition in $\mathbb{T}_R(N_1)$ needs to read and write the required links from the link places P_L. Finally, a link is consumed together with its object by the consuming transition in T_c.

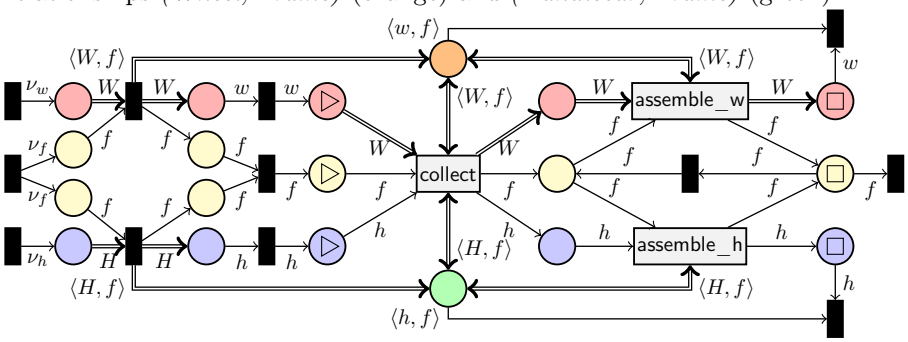
Precisely, we define the flow relations as follows:

- For a new emitting transition $t_{i,\sigma} \in T_i$, and $p \in P_b$ where $\text{color}_2(p) = \langle\sigma\rangle$, the outflow is set as $F_{out,2}(t_{i,\sigma}, p) = \langle\nu_\sigma\rangle$.
- For a former emitting transition $t_{e,\sigma} \in T_e$ and $p \in P_{\blacktriangleright}$ with $\text{color}_2(p) = \langle\sigma\rangle$, we set $F_{out,2}(t_{e,\sigma}, p) = \langle x_\sigma\rangle$ if there is a $\sigma' \in \Sigma$ such that (σ, σ') or (σ', σ) in R, otherwise $F_{out,2}(t_{e,\sigma}, p) = \langle\nu_\sigma\rangle$ as before.
- For a place in P_b which is for the one-side of the relationship, the inflow reads a single object $F_{in,2}(p_{\sigma_o,(\sigma_m,\sigma_o)}, t_{\sigma_m,\sigma_o}) = \langle x_{\sigma_o}\rangle$; while for the many-side we have $F_{in,2}(p_{\sigma_m,(\sigma_m,\sigma_o)}, t_{\sigma_m,\sigma_o}) = \langle X_{\sigma_m}\rangle$ reading a list of objects.
- Similarly, for a place in P_a for the one-side of a relationship we have $F_{out,2}(t_{\sigma_m,\sigma_o}, p_{(\sigma_m,\sigma_o),\sigma_o}) = \langle x_{\sigma_o}\rangle$ whereas for a place for the many-side we set $F_{out,2}(t_{\sigma_m,\sigma_o}, p_{(\sigma_m,\sigma_o),\sigma_m}) = \langle X_{\sigma_m}\rangle$.
- Each link creation transition $t_{\sigma_m,\sigma_o} \in T_L$ has an outflow to the respective place $p_{(\sigma_m,\sigma_o)} \in P_L$ with inscription $F_{out,2}(t_{\sigma_m,\sigma_o}, p_{(\sigma_m,\sigma_o)}) = \langle X_{\sigma_m}, x_{\sigma_o}\rangle$.
- Every former emitting transition $t_{e,\sigma} \in T_e$ has an inflow from all related $p \in P_a$ where $\text{color}_2(p) = \langle\sigma\rangle$, so that $F_{in,2}(p, t_{e,\sigma}) = \langle x_\sigma\rangle$.
- For all link places $p_{(\sigma_m,\sigma_o)} \in P_L$ and transitions $t \in T$ where there exist places $p_1, p_2 \in P$ such that $\text{color}_2(p_1) = \langle\sigma_m\rangle$ and $\text{color}_2(p_2) = \langle\sigma_o\rangle$ and both $F_{in,1}(p_1, t)$ and $F_{in,1}(p_2, t)$ are defined, we set $F_{in,2}(p_{(\sigma_m,\sigma_o)}, t) = F_{out,2}(t, p_{(\sigma_m,\sigma_o)}) = \langle X_{\sigma_m}, x_{\sigma_o}\rangle$.
- Finally, for all link places $p_{(\sigma_m,\sigma_o)} \in P_L$, an arc to the consuming transition $t_{c,\sigma_m} \in T_c$ consumes the links $F_{in,2}(p_{(\sigma_m,\sigma_o)}, t_{c,\sigma_m}) = \langle x_{\sigma_m}, x_{\sigma_o}\rangle$.

Note that for the modification of transitions from the OCPN, we assume that the OCPN is constructed from sound workflow nets. Hence, every transition has an inflow for an object type if and only if it has an outflow for it. E.g., if $p_{W,F}$ holds the tokens $(w_1, f_1), (w_2, f_1)$, then the transition `assemble_w` can only bind the frame f_1 to any of the wheels w_1 or w_2 but never to w_3.

For all other $p \in P_1$ and $t \in T_1$ we set $F_{in,2}(p, t) = F_{in,1}(p, t)$, $F_{out,2}(t, p) = F_{out,1}(t, p)$, and $\text{color}_2(p) = \text{color}_1(p)$. The new places are $P_2 = P_1 \cup P_b \cup P_L \cup P_a$ and the new transitions are $T_2 = T_1 \cup T_i \cup T_L$. Then, the result of the mapping $\mathbb{T}_R$ is the OPID $\mathbb{T}_R(\mathbb{T}_1(N)) = (\Sigma_1, P_2, T_2, F_{in,2}, F_{out,2}, \text{color}_2, \ell_2)$ where $\ell_2(t) = \tau$ for all $t \in T_i \cup T_L$ and $\ell_2(t) = \ell_1(t)$ otherwise.

Example 4 (Synchronizing OPID for bike manufacturing). The OPID is obtained from the OCPN in Example 1 by applying $\mathbb{T}_1$ and $\mathbb{T}_R$ for the stable relationships *(Wheel, Frame)* (orange) and *(Handlebar, Frame)* (green).



The OPID N_2 is again interpreted as an accepting OPID where the initial marking $M_{init,2}$ and final marking $M_{final,2}$ require all places to be empty. The OPID in Example 4 accepts L_1 because there exists a run, where the links for wheels and frames can be set as $(w_1, f_1), (w_2, f_1), (w_3, f_2), (w_4, f_2)$ in the orange link place and tokens for handlebars and frames as $(h_1, f_1), (h_2, f_2)$ in the green link place. The transitions bind objects correctly according to the links. However, the run in log L_2 is rejected, because $\langle \mathsf{collect}, \{f_3, h_3, w_5, w_6\}\rangle$ requires the links (w_5, f_3) and (w_6, f_3) but $\langle \mathsf{assemble_w}, \{f_4, w_6, w_7\}\rangle$ requires the link (w_6, f_4), which can not exist at the same time.

4.2 Equivalence for executions with stable relationships

In Section 3, we show that an OCPN N is equivalent to its OPID representation $N_1 = \mathbb{T}_1(N)$ for event log replay. The mapping $N_2 = \mathbb{T}_R(N_1)$ implements the stable relationships R into N_1. We show that a run in the OCPN N is conform to R if and only if there is an accepted corresponding run in N_2.

Theorem 2. *Let $N_1 = \mathbb{T}_1(N)$ be an equivalent OPID for an OCPN N and $N_2 = \mathbb{T}_R(N_1)$ a synchronizing OPID. There is an accepted run of N_1 of the form*

$$M_\emptyset \xrightarrow{*} M_0 \xrightarrow{(t_1,\beta_1)} M_1 \xrightarrow{(t_2,\beta_2)} \cdots \xrightarrow{(t_n,\beta_n)} M_n \xrightarrow{*} M_\emptyset$$

that conforms to the stable relationships R if and only if there is an accepted run of N_2 of the form

$$M_\emptyset \xrightarrow{*} M'_b \xrightarrow{*} M'_a \xrightarrow{*} M'_0 \xrightarrow{(t_1,\beta_1)} M'_1 \xrightarrow{(t_2,\beta_2)} \cdots \xrightarrow{(t_n,\beta_n)} M'_n \xrightarrow{*} M_\emptyset$$

such that $\forall p \in P_1 \colon M'_i(p) = M_i(p)$ for all $0 \leq i \leq n$.

While the detailed proof for Theorem 2 is spelled out in Appendix B of the extended version of the paper,[3] we outline its three main arguments: (i) Any

[3] `https://anonymous.4open.science/r/To-bind-or-not-to-bind-43DD`

accepted run in N_2 is accepted in N_1. (ii) Any accepted run in N_1 that conforms to R is accepted by N_2. (iii) Any accepted run in N_1 that not conforming to R is rejected by N_2.

(i) The emitting and link creating sequence $M_\emptyset \xrightarrow{*} M'_b \xrightarrow{*} M'_a \xrightarrow{*} M'_0$ results in a marking M'_0 that holds only tokens in the places $P_\blacktriangleright \cup P_L$. An equivalent emitting sequence $M_\emptyset \xrightarrow{*} M_0$ exists for N_1 that produces the same objects in the initial places $P_\blacktriangleright$. Any binding execution (t_i, β_i) in the run of N_2 can be replayed in a corresponding marking of N_1 as it is less restrictive. Finally, there exists an appropriate sequence $M_n \xrightarrow{*} M_\emptyset$ of only consuming transitions $t \in T_c$ that removes the same objects as $M'_n \xrightarrow{*} M_\emptyset$.

(ii) For the conforming run $M_\emptyset \xrightarrow{*} M_0$ that places objects to $P_\blacktriangleright$ in N_1, there exists a corresponding run $M_\emptyset \xrightarrow{*} M'_b \xrightarrow{*} M'_a \xrightarrow{*} M'_0$ in N_2, which places the same objects onto the places P_b in M'_b. Then, $M'_b \xrightarrow{*} M'_a$ creates all required links according to the binding executions (t_i, β_i) in the run of N_1, such that for every object o_m of type σ_m in a relationship with an object of type σ_o, links in P_L are set for all $lo(o_m, \sigma_o)$. Next, $M'_a \xrightarrow{*} M'_0$ puts the objects into $P_\blacktriangleright$ such that M'_0 holds tokens only in places $p' \in P_\blacktriangleright \cup P_L$. The created links are not changed and transitions that operate on two object types in a relationship are required to adhere to them.
If the run in N_1 is conform, it only binds an object o_m to a single object of σ_o hence, $|lo(o_m, \sigma_o)| = 1$ and there exists a link such that N_2 can accept the run as well. Finally, for $M_n \xrightarrow{*} M_\emptyset$ of only consuming transitions $t \in T_c$, $M'_n \xrightarrow{*} M_\emptyset$ can consume the same objects including their respective links.

(iii) By contradiction, we show that N_2 cannot create links that violate the conformance criteria. No object in a relationship can exist without a link. Any object o_m with $ot(o_m) = \sigma_m$ will receive exactly one link to an object of type σ_o and not more. As the bindings of any run in N_2 are restricted by the links, a non-conforming run of N_1 has no counterpart in N_2.

By extending the mapping $\mathbb{T}_1$ with stable relationships in $\mathbb{T}_R$ for a given OCPN and its relationships R, we can generate an OPID $\mathbb{T}_R(\mathbb{T}_1(N))$ that accepts an execution in N if and only if it is conforming to R.

5 Discussion and Related Work

The conceptual implication of Theorem 2 is that whenever one is interested only in executions with stable (many-to-one) relationships, OCPNs not underfitting but well-suited. In fact, for such executions an OCPN can be unambiguously encoded into a corresponding OPID where the manipulation of objects, as well as the creation of stable links, and their consequent access for synchronization, is spelled out in its full extent.

Stable relationships at the type level can be discovered from OCELs, as described in the tool TOTeM [16]. To substantiate our intuition that in many contexts OCPNs are "just fine" to describe an stable relationships in object-

Table 1. OCELs with their number of object types, all relationships (including one-to-one), stable many-to-one relationships with, and without filtering noise.

OCEL	#obj. types	#relationships	#m2o (noise=0.1)	#m2o (noise=0.0)
Logistics	7	9	6	6
Order Management	6	13	5	5
Procure to Pay	7	9	10	10
Hinge Production	12	39	35	35
Age of Empires	30	135	**87**	**67**
LRMS – O2C	9	11	**18**	**16**
LRMS – P2P	6	15	**11**	**0**
LRMS – Hiring	6	15	6	6
LRMS – Hospital	8	18	6	6

centric process , we applied an according analysis to existing benchmark logs[4]. The result in Table 1 shows that all logs contain stable many-to-one relationships.

Recall that stability breaks when the same object from the "many" side is bound to two distinct objects from the "one side" by two events. It is not possible to differentiate between a non-rigid one-to-many relationship and an intended many-to-many relationship, just by looking at the OCEL without domain knowledge. Therefore, our approach emphasizes an explicit declaration of object types and relationships, e.g., as data models. The need of enriching object-centric event data with concepts from data modeling is being increasingly recognized [8,12,17].

At the same time, stability may be broken due to the presence of noise, i.e., due to infrequent outlier events. In the table, we report how the number of stable relationships change when considering noise Some logs contain noisy violations of stable relationships. For instance, the LLM-based simulations for LRMS – O2C contain 20 stable relationships more when only focusing on the 90% most frequent observations. This relates to the approach in [21], where this type of violation is identified in a real-world use case. Our mapping to OPIDs can assist in identifying the exact executions that violate stable relationships.

The stability of object relationships is a typical (often, implicit) assumption in the literature. In the widely adopted OCEL 2.0 standard [6], links are recorded without a notion of time, assuming global existence and stability. For discovered models, such as OCPNs [4], class diagrams [16], and even simulation models [13], this yields the same notion of stable relationship introduced here.

If an input OCEL guarantees stability, a discovered OCPN, as discovered in [4], can provide guarantees on fitness and precision. Under this assumption, the mapping given in Section 4.1 shows how to directly employ these techniques to discover OPIDs. The discovery technique in in [4] has been recently extended in [19], who propose the discovery of object-centric process trees. Identifier-sound [18] OCPNs can be generated from such process trees, yet without extending on the synchronization semantics. At the same time, [20] introduces discovered silent objects that reify and synchronize arbitrary sets of objects. These silent objects may provide insights on how to extend our investigation to many-to-many relationships. The conformance metrics of fitness and precision

[4] `https://anonymous.4open.science/r/To-bind-or-not-to-bind-43DD`

have been introduced to quantify the appropriateness of an OCPN in regard to replaying a log [5], and techniques for computing alignments for OCPNs [15], have been studied. Both focus only on individual objects, and not on their mutual links. For varying links, these approaches are too liberal in identifying deviations [11], calling for richer formalisms like OPIDs, which are computationally more demanding [11]. Still, we show that conformance checking with OCPNs [5, 15] is appropriate for stable links.

As a conclusive remark, our results indicate that under the stability assumption, OCPNs and related process mining techniques provide a good reference framework. In more complex scenarios where relationships are not rigid (such as logistic processes where packages may be transferred from a vehicle to another vehicle), richer formalisms and techniques are needed instead. As presented in Table 1, executions violating stable relationships exists and needs formalisms for recognizing them. The need of dealing with object-centric event data with non-rigid relationships is in fact being increasingly recognized. For example, [12] points out that from a database perspective, links between objects might indeed change over time. [17] highlights the need of richer representations for object-centric event data, where the effects of events are explicitly tracked, enabling the identification of time intervals within which a link holds. An ongoing effort aims to consolidate these different considerations in a single standard for representing object-centric event data [8], motivating further research on the matter.

6 Conclusion

All in all, our contribution closes the gap between two representative formal models for object-centric processes: OCPNs—an underfitting model with effective and modular discovery and available conformance checking techniques, and OPIDs—a much more sophisticated model for which discovery techniques are still missing, and conformance checking techniques are computationally more demanding. Our findings indicate that when focusing on executions where many-to-one relationships are stable, that is, once a link is established it is rigidly maintained over time, there is a non-ambiguous way to relate these two models. The implications of this result can be read in two ways. In one direction, it shows that while OCPNs are, in general, underfitting, they become precise enough when describing and prescribing executions that establish stable many-to-one relationships. On the other hand, when restricting attention to this type of executions, discovery and conformance checking for OPIDs can be directly tackled through techniques natively defined for OCPNs. In the future, stability in many-to-many relations in event logs remains to be studied in detail. Furthermore, our results can serve as a promising link between object-centric process discovery and formal analysis as well as model-driven process execution.

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A Proof for Equivalent OPIDs

To prove the language equivalence of N and $\mathbb{T}_1(N)$ in general, we first define some auxiliary notions.

Given a marking $M\colon Q \to \{0,1\}$ of the OCPN N, we define the corresponding marking $\mathbb{T}_1(M)$ of the OPID $\mathbb{T}_1(N)$ as $\mathbb{T}_1(M)(p) := \{\langle o\rangle \mid o \in \mathcal{O} \wedge M(p,o) > 0\}$.

Given a binding b for an OCPN N, we define a corresponding binding $\beta := \mathbb{T}_1(b)$ for the OPID $\mathbb{T}_1(N)$ as follows: for all $\sigma \in dom(b)$, we set $\beta(v_\sigma) = b(\sigma)$ if $|b(\sigma)| = 1$, and $\beta(V_\sigma) = [b(\sigma)]$ otherwise, where $[b(\sigma)]$ denotes a list containing the elements of the set $b(\sigma)$ in an arbitrary order.

Moreover, we call a sequence of transition firings $M \xrightarrow{*} M'$ in $\mathbb{T}_1(N)$ an *emitting sequence* if it uses only emitting transitions in T_e and a *collapsing sequence* if it uses only consuming transitions in T_c.

We can then show the following equivalence statement:

Theorem 1. *Let O be a set of objects, N be an accepting OCPN, $N' = \mathbb{T}_1(N)$, and $M_\emptyset$ denote the empty marking of N'.*

There exists a run of N:

$$M_0 \xrightarrow{(t_1,b_1)} M_1 \xrightarrow{(t_2,b_2)} \cdots \xrightarrow{(t_n,b_n)} M_n \tag{1}$$

with $M_0{=}M^O_{init}$ and $M_n{=}M^O_{final}$ if and only if there is a run of N' over the set of objects O:

$$M_\emptyset \xrightarrow{*} M'_0 \xrightarrow{(t_1,\beta_1)} M'_1 \xrightarrow{(t_2,\beta_2)} \cdots \xrightarrow{(t_n,\beta_n)} M'_n \xrightarrow{*} M_\emptyset \tag{2}$$

such that $M'_i = \mathbb{T}_1(M_i)$ for all $0 \le i \le n$ and $\beta_i = \mathbb{T}_1(b_i)$ for all $1 \le i \le n$, such that $M_\emptyset \xrightarrow{*} M'_0$ is an emitting sequence, $M'_n \xrightarrow{*} M_\emptyset$ a collapsing sequence, and no emitting or collapsing transitions occur in between.

Proof. (1) First, by definition of T_e in N', there is an emitting sequence $M_\emptyset \xrightarrow{*} M'_0$ since $M'_0 = \mathbb{T}_1(M_0)$ has tokens only in places that are initial in N. Moreover, by definition of T_c there is a collapsing sequence $M'_n \xrightarrow{*} M_\emptyset$ because $M'_n = \mathbb{T}_1(M_n)$ has tokens only in places that are final in N.

Next, we show the following ($\star$): for all $1 \le i \le n$, a transition firing $M_{i-1} \xrightarrow{(t_i,b_i)} M_i$ implies $M'_{i-1} \xrightarrow{(t_i,\beta_i)} M'_i$. Note first that the binding $\beta_i = \mathbb{T}_1(b_i)$ is type-preserving since this also holds for b_i; and it is well-defined because transitions in an OCPN have at most one input place for each sort, so the transition t_i of the OPID $\mathbb{T}_1(N)$ cannot have two input flows with the same inscription. All inscriptions in $\mathbb{T}_1(N)$ are singleton tuples, so the extension $\boldsymbol{\beta}_i$ of β_i to inscriptions is given by $\boldsymbol{\beta}_i(\langle x_\sigma\rangle) = \{\langle \beta_i(x_\sigma)\rangle\}$ for variables $x_\sigma \in \Upsilon \cup X$, and by $\boldsymbol{\beta}_i(\langle X_\sigma\rangle) = \{\langle a\rangle \mid a \text{ occurs in } \beta_i(X_\sigma)\}$ for all $X_\sigma \in X_{list}$. In both cases, we have $\boldsymbol{\beta}_i(x_\sigma) = \boldsymbol{\beta}_i(X_\sigma) = \{\langle a\rangle \mid a \in b_i(\sigma)\}$.

Now, as t_i is enabled with b_i in marking M_{i-1} of N, $cons(t_i,b_i) \le M_{i-1}$ holds for $cons(t_i,b_i) = \{(p,o) \mid p \in \bullet t_i \text{ and } o \in b_i(pt(p))\}$. We have $\boldsymbol{\beta}_i(F_{in}(p,t_i)) = \{\langle o\rangle \mid o \in b_i(pt(p))\}$, and therefore $\boldsymbol{\beta}_i(F_{in}(p,t_i)) \subseteq M'_{i-1}(p)$

for all $p \in \bullet t_i$, so that t_i is enabled in marking $\mathbb{T}_1(M_{i-1})$ of N'. The resulting marking in N is $M_i = (M_{i-1} \setminus cons(t_i, b_i)) \cup prod(t, b)$, for $prod(t, b) = \{(p, o) \mid p \in t\bullet$ and $o \in b(pt(p))\}$. The equivalent marking in N' is given by $M'_i(p) = M'_{i-1}(p) \setminus \boldsymbol{\beta}_i(F_{in}(p, t_i))$ for all $p \in \bullet t_i \setminus t_i\bullet$, $M'_i(p) = M'_{i-1}(p) \cup \boldsymbol{\beta}_i(F_{out}(p, t_i))$ for all $p \in t\bullet_i \setminus \bullet t_i$, and $M'_i(p) = M'_{i-1}(p) \setminus \boldsymbol{\beta}_i(F_{in}(p, t_i)) \cup \boldsymbol{\beta}_i(F_{out}(p, t_i))$ for all $p \in \bullet t_i \cap t\bullet_i$. We have $\boldsymbol{\beta}_i(F_{in}(p, t_i)) = \boldsymbol{\beta}_i(F_{out}(p, t_i)) = \{\langle o \rangle \mid o \in b_i(pt(p))\}$, so $M'_i = \mathbb{T}_1(M_i)$. Thus $(\star)$ holds.

A simple inductive argument using $(\star)$, in addition to the observations about emitting and collapsing sequences, shows shows that the run (2) is possible.

(2) First, since $M_\emptyset \xrightarrow{*} M'_0$ is an emitting sequence from the initial marking, M'_0 has tokens only in places that are initial in N. So there is a marking M_0 of N that is a valid initial marking of N and satisfies $M'_0 = \mathbb{T}_1(M_0)$.

Next, we show the following $(\star)$: for all $1 \leq i \leq n$, given M_{i-1} which satisfies $M'_{i-1} = \mathbb{T}_1(M_{i-1})$, there are a marking M_i and a binding b_i that satisfy $M_{i-1} \xrightarrow{(t_i, b_i)} M_i$, $M'_i = \mathbb{T}_1(M_i)$, and $\beta_i = \mathbb{T}_1(b_i)$. Since $\mathbb{T}_1$ is injective on markings and bindings, M_i and b_i can actually be obtained from M'_i and β_i by applying the inverse of $\mathbb{T}_1$. As t_i is enabled in M'_{i-1}, it holds that $\boldsymbol{\beta}_i(F_{in}(p, t_i)) \subseteq M'_{i-1}(p)$ for all $p \in \bullet t_i$. Since $\boldsymbol{\beta}_i(F_{in}(p, t_i)) = \{\langle o \rangle \mid o \in b_i(pt(p))\}$ (see Item (1)), $cons(t_i, b_i) \leq M_{i-1}$ holds, so t_i is enabled in M_{i-1}. The resulting marking M'_i in N' satisfies $M'_i(p) = M'_{i-1}(p) \setminus \boldsymbol{\beta}_i(F_{in}(p, t_i))$ for all $p \in \bullet t_i \setminus t_i\bullet$, $M'_i(p) = M'_{i-1}(p) \cup \boldsymbol{\beta}_i(F_{out}(p, t_i))$ for all $p \in t\bullet_i \setminus \bullet t_i$, and $M'_i(p) = M'_{i-1}(p) \setminus \boldsymbol{\beta}_i(F_{in}(p, t_i)) \cup \boldsymbol{\beta}_i(F_{out}(p, t_i))$ for all $p \in \bullet t_i \cap t\bullet_i$. The resulting marking M_i in N satisfies $M_i = (M_{i-1} \setminus cons(t_i, b_i)) \cup prod(t, b)$. Since $\boldsymbol{\beta}_i(F_{in}(p, t_i)) = \boldsymbol{\beta}_i(F_{out}(p, t_i)) = \{\langle o \rangle \mid o \in b_i(pt(p))\}$, it holds that $M'_i = \mathbb{T}_1(M_i)$. Thus $(\star)$ holds and an inductive argument shows that the run (1) is possible.

Finally, since $M'_n \xrightarrow{*} M_\emptyset$ is a collapsing sequence to an empty marking, $M'_n = \mathbb{T}_1(M_n)$ has tokens only in places that are final in N, so M_{final} is a suitable final marking for N. □

B Proof for Synchronizing OPIDs

In Appendix A, the equivalence of an OCPN N and its OPID representation $N_1 = \mathbb{T}_R(N)$ for replaying an event log is shown. The mapping $\mathbb{T}_R$ implements the stable relationships R into N_1. We prove the correct mapping by showing three things: First, any accepted run in N_2 is also accepted in N_1. Second, any accepted run in N_1, which is conform to the stable relationships R, is accepted by N_2. Third, any accepted run in N_1, which that does not conform to the stable relationships R, is rejected by N_2.

Theorem 2. *Let $N_1 = \mathbb{T}_1(N) = (\Sigma_1, P_1, T_1, F_{in,1}, F_{out,1}, \mathrm{color}_1, \ell_1)$ be the equivalent OPID for the accepting OCPN N over the set of objects $\mathcal{O}$. Let $N_2 = \mathbb{T}_R(N_1) = (\Sigma_2, P_2, T_2, F_{in,2}, F_{out,2}, \mathrm{color}_2, \ell_2)$ for the set R of stable object type relationships, and $M_\emptyset$ denote the empty marking of N_1 and N_2.*

(1) For any accepted run of N_2:

$$M_\emptyset \xrightarrow{*} M'_0 \xrightarrow{(t_1,\beta_1)} M'_1 \xrightarrow{(t_2,\beta_2)} \cdots \xrightarrow{(t_n,\beta_n)} M'_n \xrightarrow{*} M_\emptyset$$

such that $M_\emptyset \xrightarrow{*} M'_0$ is an emitting and link creation sequence of only transitions in $t \in T_i \cup T_L \cup T_e$, $M'_n \xrightarrow{*} M_\emptyset$ is a collapsing sequence of only transitions $t \in T_c$, and no transitions $t \in T_i \cup T_L \cup T_e \cup T_c$ occur in between, there is an accepted run of N_1 of the form

$$M_\emptyset \xrightarrow{*} M_0 \xrightarrow{(t_1,\beta_1)} M_1 \xrightarrow{(t_2,\beta_2)} \cdots \xrightarrow{(t_n,\beta_n)} M_n \xrightarrow{*} M_\emptyset \tag{3}$$

such that $\forall p \in P_1 \colon M'_i(p) = M_i(p)$ for all $0 \leq i \leq n$.

(2) For an accepted run of N_1 that respects the many-to-one relationships R:

$$M_\emptyset \xrightarrow{*} M_0 \xrightarrow{(t_1,\beta_1)} M_1 \xrightarrow{(t_2,\beta_2)} \cdots \xrightarrow{(t_n,\beta_n)} M_n \xrightarrow{*} M_\emptyset$$

there is an accepted run of N_2 of the form

$$M_\emptyset \xrightarrow{*} M'_b \xrightarrow{*} M'_a \xrightarrow{*} M'_0 \xrightarrow{(t_1,\beta_1)} M'_1 \xrightarrow{(t_2,\beta_2)} \cdots \xrightarrow{(t_n,\beta_n)} M'_n \xrightarrow{*} M_\emptyset \tag{4}$$

such that $\forall p \in P_1 \colon M'_i(p) = M_i(p)$ for all $0 \leq i \leq n$.

(3) For an accepted run of N_1 that does not respect the stable relationships R:

$$M_\emptyset \xrightarrow{*} M_0 \xrightarrow{(t_1,\beta_1)} M_1 \xrightarrow{(t_2,\beta_2)} \cdots \xrightarrow{(t_n,\beta_n)} M_n \xrightarrow{*} M_\emptyset$$

there can be no accepted run of N_2 of the form

$$M_\emptyset \xrightarrow{*} M'_0 \xrightarrow{(t_1,\beta_1)} M'_1 \xrightarrow{(t_2,\beta_2)} \cdots \xrightarrow{(t_n,\beta_n)} M'_n \xrightarrow{*} M_\emptyset \tag{5}$$

such that $\forall p \in P_1 \colon M'_i(p) = M_i(p)$ for all $0 \leq i \leq n$.

Proof. (1) $M_\emptyset \xrightarrow{*} M'_0$ in N_2 produces a set of objects through T_i, creates links through T_L, and the transitions $t \in T_e$ populate the play places $p \in P_\blacktriangleright$. In M'_0, only the places $P_\blacktriangleright$ and P_L hold tokens, i.e. $M'_0(p) = \emptyset$ for all $p \in P_2 \setminus (P_\blacktriangleright \cup P_L)$, because no transition $t \in T \setminus (T_e \cup T_L$ can remove the remaining tokens from any place $p \in P_b \cup P_a$, as for all those places and transitions it holds that $F_{in}(p,t) = \bot$, and the execution would not be accepted for the final marking $M_\emptyset$.

For N_1, there exists a sequence $M_\emptyset \xrightarrow{*} M_0$ that can generate the same marking for the play places $p \in P_\blacktriangleright \subset P_1$ by firing the appropriate emitting transition for every object. Indeed, for all objects and places $p' \in P_\blacktriangleright \subset P_2$, where $o \in M'_0(p')$ and $ot(o) = \sigma$, there exists an emitting transition $t \in T_e \subset T_1$ with an outflow to the according place $p \in P_\blacktriangleright \subset P_1$ with $\text{color}_2(p) = \sigma$ where $F_{out,1}(t,p) = \langle \nu_\sigma \rangle$. Therefore, the emitting transition can produce o onto p. Hence, given the marking M'_0 in N_2, N_1 can produce an equivalent marking M_0 for all places $p \in P_1$.

Next we show the following ($\star$): For all $1 \leq i \leq n$, a transition firing $M'_{i-1} \xrightarrow{(t_i,\beta'_i)} M'_i$ implies the existence of a firing $M_{i-1} \xrightarrow{(t_i,\beta_i)} M_i$ where for all $p \in P_1$ it holds that $M_{i-1}(p) = M'_{i-1}(p)$. For all transitions $t' \in T_1 \subset T_2$ of N_2 and $t \in T_1$ of N_1 and places $p' \in P_1 \subset P_2$ of N_2 and $p \in P_1$ of N_1, the inflow and outflow are the same for both nets $F_{in,1}(t,p) = F_{in,2}(t',p')$ and $F_{out,1}(t,p) = F_{out,2}(t',p')$. Hence if the transition (t_i, β_i) is enabled in marking M'_{i-1} of N_2 then it is is also enabled in M_{i-1} of N_1, and the resulting markings satisfy $M_i(p) = M'_i(p)$ for all places $p \in P_1$.
In M'_n, only the places $P_\blacksquare$ and P_L hold tokens, so $M'_n(p) = \emptyset$ for all $p \in P_2 \setminus (P_\blacksquare \cup P_L)$. Otherwise, no transition $t \in T_c \subset T_2$ can remove the remaining tokens from any place p, as for all those places and transitions it holds that $F_{in}(p,t) = \bot$, and the execution would not be accepted for the final marking $M_\emptyset$. Consider the collapsing sequence $M'_n \xrightarrow{*} M_\emptyset$ in N_2 of only transitions $t' \in T_c$ and a marking M_n in N_1, where $M_n(p) = M'_n(p)$ for all places $p \in P_1$. There exists a collapsing sequence in N_1 consisting of only consuming transitions $t \in T_c$, resulting in $M_\emptyset$ as $F_{in,1}(p,t) = \langle x_{\mathrm{color}_2(p)} \rangle$ and $F_{out,1}(t,p') = \bot$ for any $p' \in P_1$.
A simple inductive argument using ($\star$), and the observations about the emitting, link creating, and the collapsing sequences, show that the run (3) is possible.

(2) For the emitting sequence $M_\emptyset \xrightarrow{*} M_0$ in N_1 and the set of relations R, there is an emitting sequence $M_\emptyset \xrightarrow{*} M'_b \xrightarrow{*} M'_a \xrightarrow{*} M'_0$ in N_2 that sets the links correctly, as follows. Let $S \subseteq \Sigma$ be the set of object types that occur in R.
   - The sequence $M_\emptyset \xrightarrow{*} M'_b$ creates objects whose type occurs in R. To that end, it uses transitions $t_i \in T_i$ to obtain a marking M'_b where for all places $p_b \in P_b$ with $\mathrm{color}_2(p_b) = \langle \sigma \rangle$ for $\sigma \in S$, and the unique place $p \in P_\blacktriangleright$ such that $\mathrm{color}_2(p_b) = \mathrm{color}_2(p)$, it holds that $M_b(p_b) = M_0(p)$ because $F_{out,2}(t_i, p_b) = \langle \nu_\sigma \rangle$.
   - The sequence $M'_b \xrightarrow{*} M'_a$ uses only transitions $t_{\sigma_m,\sigma_o} \in T_L$, with the aim to populate P_L with the required links. For every relationship $(\sigma_m, \sigma_o) \in R$, there exist two places $p_{\sigma_m,(\sigma_m,\sigma_o)}, p_{\sigma_o,(\sigma_m,\sigma_o)} \in P_b$, one transition $t_{(\sigma_m,\sigma_o)} \in T_L$, two places $p_{(\sigma_m,\sigma_o),\sigma_m}, p_{(\sigma_m,\sigma_o),\sigma_o} \in P_a$, and one place $p_{(\sigma_m,\sigma_o)} \in P_L$ that satisfy $F_{in,2}(p_{\sigma_m,(\sigma_m,\sigma_o)}, t_{(\sigma_m,\sigma_o)}) = F_{out,2}(t_{(\sigma_m,\sigma_o)}, p_{\sigma_m,(\sigma_m,\sigma_o)}) = \langle X_{\sigma_m} \rangle$ (resp. $F_{in,2}(p_{\sigma_o,(\sigma_m,\sigma_o)}, t_{(\sigma_m,\sigma_o)}) = F_{out,2}(t_{(\sigma_m,\sigma_o)}, p_{\sigma_o,(\sigma_m,\sigma_o)}) = \langle x_{\sigma_o} \rangle$), as well as $F_{out,2}(t_{\sigma_m,\sigma_o}, p_{(\sigma_m,\sigma_o)}) = \langle X_{\sigma_m}, x_{\sigma_o} \rangle$. For every object o, if $ot(o) = \sigma_m$ (resp. $ot(o) = \sigma_o$), then $\langle o \rangle \in M'_b(p_{\sigma_m,(\sigma_m,\sigma_o)})$ (resp. $\langle o \rangle \in M'_b(p_{\sigma_m,(\sigma_m,\sigma_o)})$).
     For every transition $t_{\sigma_m,\sigma_o} \in T_L$ and every object $o_o \in \mathcal{O}$ with $ot(o_o) = \sigma_o$, a binding β_{o_o,σ_m} can be selected that creates all links of this type that are required later. Let O_m be the set of all objects of type σ_m that occur with o_o in the run of N_1, i.e., all objects $o_m \in \mathcal{O}$ such that $ot(o_m) = \sigma_m$ and $\{o_o, o_m\} \subseteq codom(\beta_i)$. Then $\beta_{o_o,\sigma_m}(x_{\sigma_o}) = o_o$ and $\beta_{o_o,\sigma_m}(X_{\sigma_m}) = [O_m]$. As all required objects are present in the according places of P_b, $(t_{\sigma_m,\sigma_o}, \beta_{o_o,\sigma_m})$ can fire. Every object o_o and $o_m \in O_m$ is only considered once by a transition t_{σ_m,σ_o},

as $F_{out,2}(t_{\sigma_m,\sigma_o}, p_{\sigma_m,(\sigma_m,\sigma_o)}) = F_{out,2}(t_{\sigma_m,\sigma_o}, p_{\sigma_o,(\sigma_m,\sigma_o)}) = \bot$. Because the run in N_1 satisfies the relationship $(\sigma_m, \sigma_o) \in R$, for all bindings β_i it holds that if $o_m \in \beta_i(x_{\sigma_m})$ and an object $o'_o \in \beta_i(x_{\sigma_m})$ of type $ot(o_o) = \sigma_o$, then $o'_o = o_o$. Therefore, for all $p_{(\sigma_m,\sigma_o)} \in P_L$ it holds that for every pair of links $(o_m, o_o), (o'_m, o'_o) \in M'_a(p_{(\sigma_m,\sigma_o)})$ if $o_m = o'_m$ then $o_o = o'_o$. All links in the places P_L are set correctly for the run in N_1.
For all transitions and bindings (t_i, β_i) in the run, the marking of all $p_{(\sigma_m,\sigma_o)} \in P_L$ is not changed as $F_{in,2}(p_{(\sigma_m,\sigma_o)}, t) = F_{out,2}(t, p_{(\sigma_m,\sigma_o)})$ for all $t \in T \setminus (T_L \cup T_c)$. All relationships are stable.

– The sequence $M'_a \xrightarrow{*} M'_0$ fires transitions to populate $P_{\blacktriangleright}$ for every object o in the run: Let $ot(o) = \sigma$. First, if $\sigma \notin S$, there is a flow from the according $t_\sigma \in T_e$ to a place $p \in P_{\blacktriangleright}$ with $F_{out,2}(t_\sigma, p) = \langle \nu_\sigma \rangle$. Second, if $ot(o) \in S$, there is a place $p_a \in P_a$ with $\text{color}_2(p_a) = \langle \sigma \rangle$ that holds a token $\langle o \rangle$. Hence, a binding $\beta(x_\sigma) = o$ exists for all $F_{in,2}(p_a, t_e) = \langle x_\sigma \rangle$, resulting in a marking where $\langle o \rangle \in M'_0(p)$ because $F_{out,2}(t_e, p) = \langle x_\sigma \rangle$.

Next we show the following $(\star)$: For all $1 \leq i \leq n$, a transition firing $M_{i-1} \xrightarrow{(t_i,\beta_i)} M_i$ where for all $p \in P_1$ it holds that $M_{i-1}(p) = M'_{i-1}(p)$, implies the existence of a firing $M'_{i-1} \xrightarrow{(t_i,\beta_i)} M'_i$, such that that $M_i(p) = M'_i(p)$. For all transitions $t \in T_1 \setminus (T_e \cup T_c) = T_2 \setminus (T_i \cup T_L \cup T_e \cup T_c)$ and all places $p \in P_2 \setminus (P_L \cup P_b \cup P_a) = P_1$, it holds that all arc inscriptions are singleton tuples such that $F_{out,2}(t, p) = F_{out,1}(t, p)$, $F_{in,2}(p, t) = F_{in,1}(p, t)$. If the binding β_i considers objects $o_m \in O_m$ with $ot(o_m) = \sigma_m$ and o_o with $ot(o_o) = \sigma_o$ of two types that are in a relationship $(\sigma_m, \sigma_o) \in R$, there is a place $p_{\sigma_m,\sigma_o} \in P_L \subset P_2$ where $O_m \times \{o_o\} \subseteq M'_0(p_{\sigma_m,\sigma_o})$. Because the relationship is stable, we also have $O_m \times \{o_o\} \subseteq M'_{i-1}(p_{\sigma_m,\sigma_o})$. Therefore, for the binding β_i from the run of N_1 can also be used in the run of N_2. As inflow and outflow are equal for all t_i and places $p \in (P_1 \cup P_L) \subset P_2$, for the resulting marking satisfies $M_i(p) = M'_i(p)$.
In M_n, only the places $P_{\blacksquare}$ hold tokens, such that $M_n(p) = \emptyset$ for all $p \in P_1 \setminus P_{\blacksquare}$. Otherwise, no transition $t \in T_c \subset T_1$ could remove the remaining tokens from any place $p \in P_1 \setminus P_{\blacksquare}$, because $F_{in}(p, t) = \bot$, and $M_\emptyset$ could not be reached. For the collapsing sequence $M_n \xrightarrow{*} M_\emptyset$, there exists an equal collapsing sequence $M'_n \xrightarrow{*} M_\emptyset$ of transitions $t \in T_c \subset T_2$. For every object $o_m \in \mathcal{O}$ of type $ot(o_m) = \sigma_m$ and every relationship (σ_m, σ_o), there exists a place $p_{\sigma_m,\sigma_o} \in P_L$ with exactly one token $(o_m, o_o) \in M'_n(p_{\sigma_m,\sigma_o})$. There exists a final place $p \in P_{\blacksquare}$ with $o_m \in M'_n(p)$, and $F_{in,2}(p, t_e) = \langle x_{\sigma_m} \rangle$ and the link place with $F_{in,2}(p_{\sigma_m,\sigma_o}, t_e) = \langle x_{\sigma_m}, x_{\sigma_o} \rangle$. Hence, a binding $\beta'(\langle x_{\sigma_m}, x_{\sigma_o} \rangle) = \{o_m, o_o\}$ exists for every o_m and is enabled. Therefore, all objects can be removed with their many-to-one links.
A simple inductive argument using $(\star)$, and the observations about the emitting and link creating run, and the collapsing run, shows shows that the total run (4) is possible.

(3) A run in N_1 is not conform to a stable relationship $(\sigma_m, \sigma_o) \in R$, if there exist objects $o_m, o_o \in \mathcal{O}$, where $ot(o_m) = \sigma_m$ and $ot(o_o) = \sigma_o$, that contradict the condition $|lo(o_m, \sigma_o)| = 1$ and $lo(o_o, \sigma_m) \neq \emptyset$. A contradiction for N_1 is posed

by any of the following three options: (i) $lo(o_m, \sigma_o) = \emptyset$, (ii) $|lo(o_m, \sigma_o)| > 1$, or (iii) $lo(o_o, \sigma_m) = \emptyset$.
All transitions $t \in T_1 \subset T_2$ that operate both object types σ_m and σ_o, also read from the link place for (σ_m, σ_o) as $F_{in,2}(p_{\sigma_m,\sigma_o}, t) = F_{out,2}(t, p_{\sigma_m,\sigma_o}) = \langle X_{\sigma_m}, x_{\sigma_o} \rangle$. Any binding β_i for the transition t operating on σ_m and σ_o is bound to the link place p_{σ_m,σ_o}. A binding β_i including o_m and o_o is only enabled in the marking M'_i if $\langle o_m, o_o \rangle \in M'_i(p_{\sigma_m,\sigma_o})$. $lo()$ is defined on the bindings β_i of the run $M'_0 \xrightarrow{*} M'_n$.
Therefore, we show that the links for (σ_m, σ_o) can not be constructed in any way such that any of the contradictions (i-iii) are possible to hold in an accepted run of N_2. Note that every object o_m and o_o is considered exactly once for a transition that creates links for the relationship (σ_m, σ_o) in N_2 and during the run $M'_0 \xrightarrow{*} M'_n$, all links in p_{σ_m,σ_o} are stable (cf. item (2)). Hence, the following arguments hold true:

(i) To satisfy $lo(o_m, \sigma_o) = \emptyset$, there must exist no binding that operates on o_m and any o_o of $ot(o_o) = \sigma_o$. For any accepted run, o_m must participate in the link creation and there exists an object o_o such that p_{σ_m,σ_o} holds $\langle o_m, o_o \rangle$. A run in N_2 can only be accepted if any binding over o_m and an object of type σ_o binds o_m to the appropriate o_o. Hence, $lo(o_m, \sigma_o) \neq \emptyset$

(ii) To satisfy $|lo(o_m, \sigma_o)| > 1$, bindings must exist that correlate o_m to two objects of type σ_o. However, during link creation, any o_m can be linked to only one object of the type $ot(o_o) = \sigma_o$ such that in any marking only one link exists $|\{\langle o'_m, o'_o \rangle \in M'_i(p_{\sigma_m,\sigma_o}) \mid o_m = o'_m\}| = 1$. Any two bindings that consider o_m and any object o_o must read from the place p_{σ_m,σ_o} and bind objects according to the links. Therefore, both must bind the same o_o. Hence, $|lo(o_m, \sigma_o)| \leq 1$.

(iii) Analog to item (i), $lo(o_o, \sigma_m) = \emptyset$ is not possible as o_o must participate in the link creation to be in an accepted run. As all bindings over o_o and any objects of type σ_m must adhere to the links. Hence $lo(o_o, \sigma_m) \neq \emptyset$

By showing that all three contradictions must be false, the theorem holds true. No run 5 exists for N_2 that creates links for a non-conform execution. □